

# JTF003-EC — Initial analysis report

2026-07-16

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## Libraries

```
library(lme4)
library(kableExtra)
library(tidyverse)
library(patchwork)
library(sjPlot)
library(gt)
```

## Data

The data is already been cleaned of Test responses

```
d <- read_csv("all_withNorway_noTest_countryFill_raw.csv")
```

However, a couple more issues needs to be checked.

## Bilendi issues

Bilendi specific: - For the following Bilendi data collection, only the cases labelled as “Complete” are “validated” (paid). They passed the attention check and completed > the study fully - Complete indicates if they are paid or not for this sample - Remember that we use a time requirement [like took it “too fast”] Zimbabwe and South Africa had less questions and then Egypt and Kuwait had a few questions removed . For the Zimbabwe, South Africa and Cameroon samples, we have removed the following 38 questions on the 31/05, before collecting more data (Note: So these participants should not be excluded based on non-completion because they did not have these measures.). The measures that were excluded are the following: ● Conspiracy scale (6 questions) ● Actively open-minded thinking (14 questions) ● Intellectual Humility (10 Questions) ● Centrality of Religiosity Scale (8 Questions)

However, I have checked and it does not seem that the 31/05 is a sharp transition?

A different approach is to check the demographics questions that were asked at the end, and use them as evidence for completion?

```
# All demographic variables, primarily for inspection
demographic_vars <- c(
  "Age",
  "Gender", "Gender_4_TEXT",
  "Edu",
  paste0("Employment_", 1:8),
  "Employment_-999",
  "Employment_8_TEXT",
  "Student",
  "Religion",
  "Religion_spec",
  "Religion_spec_8_TEXT",
  "Country_Ori_YN",
  "Country_Ori",
```

```

"Country_Res_YN",
"Country_Res",
"native_lang",
"lang_fluency"
)

# Early demographic questions: evidence that the participant
# at least started the demographic section
early_demo_vars <- c(
  "Age",
  "Gender",
  "Edu"
)

# Late demographic questions: stronger evidence that the participant
# reached the end of the survey
late_demo_vars <- c(
  "Country_Ori_YN",
  "Country_Res_YN",
  "native_lang",
  "lang_fluency"
)

# Diagnostic table
bilendi_completion_check <- d %>%
  mutate(
    across(
      any_of(demographic_vars),
      ~ if (is.character(.x)) na_if(trimws(.x), "") else .x)) %>%
  mutate(
    n_early_demo_answered = rowSums(
      !is.na(across(all_of(early_demo_vars)))
    ),
    n_late_demo_answered = rowSums(
      !is.na(across(all_of(late_demo_vars)))
    ),
    # At least one of Age, Gender, or Education was answered
    started_demographics = n_early_demo_answered > 0,
    # At least one of the final demographic questions was answered
    reached_end_demographics = n_late_demo_answered > 0
  )

```

From the summary below we see that there are a number of participants in the Bilendi data (identified with `fetch_id==4`) who are not marked as Finished, but who reached the end of demographic questions:

```

bilendi_completion_check %>%
  filter(fetch_id == 4) %>%
  count(

```

```

    Finished,
    started_demographics,
    reached_end_demographics,
    name = "N"
  ) %>%
  arrange(
    Finished,
    desc(reached_end_demographics),
    desc(started_demographics)
  )

```

```

# A tibble: 5 x 4
  Finished started_demographics reached_end_demographics    N
  <lgl>    <lgl>                <lgl>                <int>
1 FALSE   TRUE                TRUE                203
2 FALSE   TRUE                FALSE                 2
3 FALSE   FALSE               FALSE                777
4 TRUE    TRUE                TRUE                2131
5 TRUE    FALSE               FALSE                 63

```

We can correct these

```

d <- bilendi_completion_check %>%
  mutate(
    Finished_original = Finished,

    Finished_corrected = case_when(
      fetch_id == 4 &
        !coalesce(Finished, FALSE) &
        reached_end_demographics ~ TRUE,

      TRUE ~ coalesce(Finished, FALSE)
    )
  )

# check how many have Finished status changed
d %>%
  count(
    Finished_original,
    Finished_corrected,
    name = "N"
  )

```

```

# A tibble: 3 x 3
  Finished_original Finished_corrected    N
  <lgl>            <lgl>                <int>
1 FALSE           FALSE                6325
2 FALSE           TRUE                 203
3 TRUE            TRUE                17559

```

## Exclusions

Exclusion criteria as stated in the [preregistration](#):

We will exclude participants who did not complete the study, failed an attention check question, failed to pass the Captcha test (i.e., their Captcha score is equal to or lower than 0.5), or are not permanent residents of the geopolitical region where data collection is taking place.

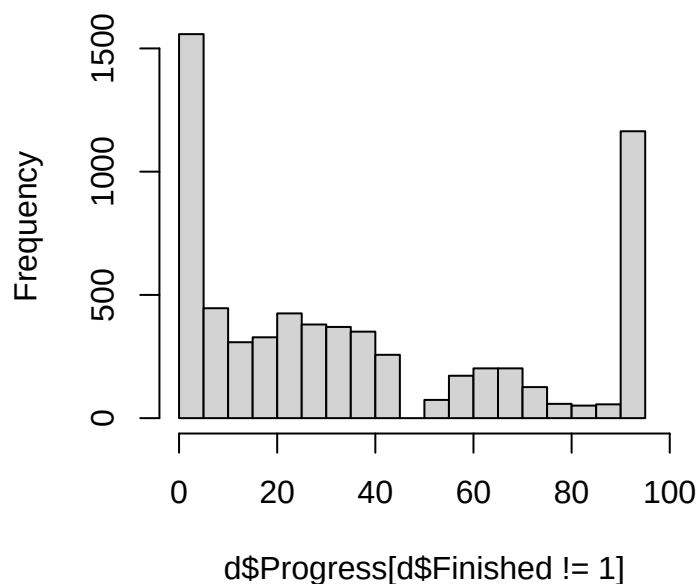
Table 1: Each line indicates how many participants failed each criterion (applied sequentially).

| Exclusion Criterion     | Number Excluded | Fraction Excluded | Percentage Excluded |
|-------------------------|-----------------|-------------------|---------------------|
| Finished == 1           | 6325            | 0.2625898         | 26.258978           |
| Q_RecaptchaScore >= 0.5 | 938             | 0.0389422         | 3.894217            |
| attention check == 0    | 355             | 0.0147382         | 1.473824            |
| Country_Res_YN == Yes   | 796             | 0.0330469         | 3.304687            |
| Total                   | 7820            | 0.3246565         | 32.465645           |

The incomplete response have quite a wide range of progress levels

```
hist(d$Progress[d$Finished!=1], xlim=c(0,100))
```

**Histogram of d\$Progress[d\$Finished != 1]**



## Countries/regions & collectivistic/individualistic categorisation

Countries included in the dataset, & number of responses per country

```
table(d$Country)
```

|               |             |                |                 |                |
|---------------|-------------|----------------|-----------------|----------------|
| Albania       | Australia   | Austria        | Bangladesh      | Belgium        |
| 17            | 410         | 287            | 89              | 92             |
| Brazil        | Bulgaria    | Cameroon       | Canada          | Chile          |
| 98            | 352         | 108            | 278             | 197            |
| China         | Colombia    | Czech Republic | Ecuador         | Egypt          |
| 136           | 372         | 158            | 2               | 183            |
| Ethiopia      | France      | Germany        | Ghana           | Greece         |
| 109           | 166         | 373            | 110             | 273            |
| Hong Kong     | Hungary     | India          | Indonesia       | Italy          |
| 70            | 226         | 275            | 123             | 896            |
| Kenya         | Kuwait      | Malaysia       | Mexico          | Morocco        |
| 143           | 169         | 349            | 122             | 96             |
| Netherlands   | New Zealand | Nigeria        | North Macedonia | Norway         |
| 383           | 104         | 144            | 186             | 79             |
| Oman          | Pakistan    | Philippines    | Poland          | Portugal       |
| 40            | 257         | 414            | 452             | 219            |
| Russia        | Serbia      | Sierra Leone   | Singapore       | Slovakia       |
| 201           | 220         | 146            | 176             | 236            |
| South Africa  | Spain       | Switzerland    | Turkey          | United Kingdom |
| 133           | 1449        | 97             | 1468            | 843            |
| United States | Vietnam     | Zimbabwe       |                 |                |
| 1729          | 97          | 81             |                 |                |

Remove participants who still have NA Country, representing 0.25% of the data.

```
d <- d %>%
  filter(!is.na(Country))
```

Add collectivistic/individualistic label

```
d <- d %>%
  mutate(culture = case_when(
    Country %in% c("Argentina", "Albania", "Australia", "Austria", "Belgium", "Brazil",
      "Bulgaria", "Canada", "Chile", "China", "Costa Rica", "Croatia",
      "Czech Republic", "France", "Germany", "Greece", "Hong Kong",
      "Hungary", "Israel", "Italy", "Japan", "Malaysia", "Netherlands",
      "New Zealand", "Norway", "Poland", "Portugal", "Russia",
      "Serbia", "Singapore", "Slovakia", "Slovenia", "South Africa",
      "Spain", "Sweden", "Switzerland", "Taiwan", "United Kingdom",
      "United States") ~ "Individualistic",
    Country %in% c("Bangladesh", "Cameroon", "Colombia", "Ecuador", "Ethiopia",
      "Egypt",
      "Ghana", "India", "Indonesia", "Iran", "Kenya",
      "Kuwait", "Malawi", "Mexico", "Morocco", "Namibia", "Nigeria",
      "North Macedonia",
      "Pakistan", "Peru", "Philippines", "Saudi Arabia", "Oman",
      "Senegal", "Sierra Leone", "Sri Lanka", "Turkey", "Vietnam", "Zimbabwe") ~
      "Collectivistic",
    TRUE ~ "Unknown"
  ))
```

```
# how many per category?
table(d$culture)
```

```
Collectivistic Individualistic
      4734          10699
```

## Demographic summary

Table 2: Demographic Information of Participants

| Demographic                | Value         |
|----------------------------|---------------|
| Average age (Std.)         | 26.20 (10.74) |
| Age Range                  | 18 to 99      |
| N. males                   | 4727          |
| N. female                  | 10350         |
| N. non binary              | 143           |
| N. prefer not to respond   | 152           |
| N. prefer to self-describe | 61            |

Split by cultural category:

Table 3: Demographic Information of Participants by Cultural Classification

| Demographic               | Individualistic | Collectivistic |
|---------------------------|-----------------|----------------|
| N participants            | 10699           | 4734           |
| Average age (SD)          | 26.79 (11.42)   | 24.86 (8.88)   |
| Age range                 | 18 to 99        | 18 to 97       |
| N males                   | 3135            | 1592           |
| N females                 | 7317            | 3033           |
| N non-binary              | 121             | 22             |
| N prefer not to respond   | 89              | 63             |
| N prefer to self-describe | 37              | 24             |

Table 4: Level of education by cultural group.

| Education level                                                    | Collectivistic, N (%) | Individualistic, N (%) |
|--------------------------------------------------------------------|-----------------------|------------------------|
| Less than a high school or secondary school degree (or equivalent) | 34 (0.7%)             | 120 (1.1%)             |
| High school or secondary school degree (or equivalent)             | 2,212 (46.7%)         | 6,582 (61.5%)          |
| Bachelor's degree (or equivalent)                                  | 1,838 (38.8%)         | 2,567 (24.0%)          |
| Postgraduate degree (MA, PhD, JD, etc.)                            | 556 (11.7%)           | 1,331 (12.4%)          |
| Prefer not to respond                                              | 94 (2.0%)             | 99 (0.9%)              |

## Data quality checks

### Use of external resources

Table 5: Percentage and number of responses reporting use of external resources, split by problem and condition.

| problem   | Baseline         | Feedback        | Justification    | Feedback + justification |
|-----------|------------------|-----------------|------------------|--------------------------|
| Problem 1 | 6.60% (N = 259)  | 6.79% (N = 267) | 7.65% (N = 288)  | 7.06% (N = 269)          |
| Problem 2 | 6.90% (N = 271)  | 6.41% (N = 252) | 8.21% (N = 309)  | 6.75% (N = 257)          |
| Problem 3 | 10.24% (N = 402) | 9.58% (N = 377) | 11.66% (N = 439) | 11.29% (N = 430)         |
| Problem 4 | 8.97% (N = 352)  | 8.67% (N = 341) | 11.16% (N = 420) | 10.37% (N = 395)         |
| Problem 5 | 9.53% (N = 374)  | 9.94% (N = 391) | 11.29% (N = 425) | 11.76% (N = 448)         |
| Problem 6 | 10.80% (N = 424) | 9.20% (N = 362) | 12.83% (N = 483) | 12.52% (N = 477)         |

## Familiarity

Table 6: Percentage and number of participants reporting familiarity, split by problem and condition.

| problem   | Baseline          | Feedback          | Justification     | Feedback + justification |
|-----------|-------------------|-------------------|-------------------|--------------------------|
| Problem 1 | 50.39% (N = 1978) | 52.75% (N = 2075) | 52.28% (N = 1968) | 54.96% (N = 2094)        |
| Problem 2 | 45.10% (N = 1770) | 46.34% (N = 1823) | 45.35% (N = 1707) | 46.61% (N = 1776)        |
| Problem 3 | 15.90% (N = 624)  | 14.36% (N = 565)  | 16.74% (N = 630)  | 16.22% (N = 618)         |
| Problem 4 | 31.16% (N = 1223) | 30.81% (N = 1212) | 31.72% (N = 1194) | 31.36% (N = 1195)        |
| Problem 5 | 26.06% (N = 1023) | 25.37% (N = 998)  | 27.42% (N = 1032) | 26.98% (N = 1028)        |
| Problem 6 | 23.01% (N = 903)  | 24.02% (N = 945)  | 24.34% (N = 916)  | 24.51% (N = 934)         |

### Language background of participants

| Response                             | N     | %    |
|--------------------------------------|-------|------|
| Fluency among non-native speakers    |       |      |
| Prefer not to respond                | 21    | 1.0  |
| Beginner                             | 46    | 2.1  |
| Intermediate                         | 413   | 19.1 |
| Advanced                             | 973   | 45.0 |
| Expert/Proficient                    | 710   | 32.8 |
| Native speaker of the study language |       |      |
| Yes                                  | 12973 | 85.0 |
| No                                   | 2164  | 14.2 |
| Prefer not to respond                | 123   | 0.8  |

Fluency was assessed only among participants who reported that the study language was not their native language.

### Language background

## Data transformation

We transform the data into a long format:

```
d <- d %>%
  mutate(participant = row_number())

d_crt_long <- d %>%
  # ID + cols matching 'crt#_i' or 'crt#_r'
  select(participant, Country, culture, justification, feedback, matches('^crt[1-6]_[ir]$')) %>%
  pivot_longer(
    cols = matches('^crt[1-6]_[ir]$'),
    names_to = 'item',
    values_to = 'response'
  ) %>%
  extract( # Extract 'problem' number and 'condition' from 'item'
    item,
    into = c('crt', 'problem', 'condition'),
    regex = '(crt)([1-6])_([ir])'
  ) %>%
  mutate(
    problem = as.integer(problem),
    condition = recode(condition, 'i' = 'intuitive', 'r' = 'reflexive')
  ) %>%
  select(participant, problem, condition, response, justification, feedback, Country, culture)

# Pivot familiarity data (one value per participant-problem)
d_fam_long <- d %>%
  pivot_longer(
    cols = matches("^familiarity_[1-6]$"),
    names_to = "fam_item",
    values_to = "familiarity"
  ) %>%
  mutate(problem = as.integer(str_extract(fam_item, "[1-6]"))) %>%
  select(participant, problem, familiarity)

# Pivot ext_resources_use data (one value per participant-problem)
d_ext_long <- d %>%
  pivot_longer(
    cols = matches("^ext_resources_use_[1-6]$"),
    names_to = "ext_item",
    values_to = "ext_resources_use"
  ) %>%
  mutate(problem = as.integer(str_extract(ext_item, "[1-6]"))) %>%
  select(participant, problem, ext_resources_use)

# Pivot dot_crt responses (one value per participant-problem)
d_dot_long <- d %>%
  pivot_longer(
    cols = matches("^dot_crt[1-6]$"),
    names_to = "dot_item",
    values_to = "dot_response"
  ) %>%
  mutate(problem = as.integer(str_extract(dot_item, "[1-6]"))) %>%
```

```

select(participant, problem, dot_response)

# correct answers to dot problems (double checked on Qualtrics) *** MATTEO: TO CHECK
dot_correct <- c(3,2,4,2,1, 3)
d_dot_long <- d_dot_long %>%
  mutate(dot_accuracy = ifelse(dot_response==dot_correct[d_dot_long$problem], 1, 0))

# Join everything together
d_long <- d_crt_long %>%
  left_join(d_fam_long, by = c("participant","problem")) %>%
  left_join(d_ext_long, by = c("participant","problem")) %>%
  left_join(d_dot_long, by = c("participant","problem")) %>%
  # Now each (participant, problem) has two rows (for each phase),
  # and familiarity and ext_resources_use are duplicated for both.
  select(participant, response, justification, feedback, problem,
    condition, problem, familiarity, ext_resources_use, dot_accuracy,
    Country, culture)

# check output
str(d_long)

```

```

tibble [185,196 x 11] (S3: tbl_df/tbl/data.frame)
 $ participant      : int [1:185196] 1 1 1 1 1 1 1 1 1 1 ...
 $ response         : num [1:185196] 2 NA 3 2 2 3 1 1 4 2 ...
 $ justification    : chr [1:185196] "J" "J" "J" "J" ...
 $ feedback         : chr [1:185196] "NF" "NF" "NF" "NF" ...
 $ problem          : int [1:185196] 1 2 3 4 5 6 1 2 3 4 ...
 $ condition        : chr [1:185196] "intuitive" "intuitive" "intuitive" "intuitive" ...
 $ familiarity      : chr [1:185196] "Yes" "No" "No" "No" ...
 $ ext_resources_use: chr [1:185196] "No" "No" "No" "No" ...
 $ dot_accuracy     : num [1:185196] 1 1 1 0 1 1 1 1 1 0 ...
 $ Country          : chr [1:185196] "Canada" "Canada" "Canada" "Canada" ...
 $ culture          : chr [1:185196] "Individualistic" "Individualistic" "Individualistic" "Individualistic" ...

```

Next we add response accuracy (the response option “1” was always the correct one)

```

# # change any -99 to NA
# d_long$response <- ifelse(d_long$response== -99, NA, d_long$response)

# compute accuracy
d_long$accuracy <- ifelse(!is.na(d_long$response),
  ifelse(d_long$response==1,1,0),
  d_long$response)

```

## Raw mean accuracy

```
aggregate(accuracy ~ I(paste(justification, feedback)) + condition,  
          data=d_long, mean)
```

|   | I(paste(justification, feedback)) | condition | accuracy  |
|---|-----------------------------------|-----------|-----------|
| 1 | J F                               | intuitive | 0.2974131 |
| 2 | J NF                              | intuitive | 0.2961667 |
| 3 | NJ F                              | intuitive | 0.3044707 |
| 4 | NJ NF                             | intuitive | 0.3033570 |
| 5 | J F                               | reflexive | 0.6813211 |
| 6 | J NF                              | reflexive | 0.5853259 |
| 7 | NJ F                              | reflexive | 0.6094306 |
| 8 | NJ NF                             | reflexive | 0.5150318 |

Now removing responses with familiarity and/or external resources use

```
aggregate(accuracy ~ I(paste(justification, feedback)) + condition,  
          data=d_long[d_long$familiarity=="No" & d_long$ext_resources_use=="No",],  
          mean)
```

|   | I(paste(justification, feedback)) | condition | accuracy  |
|---|-----------------------------------|-----------|-----------|
| 1 | J F                               | intuitive | 0.2020897 |
| 2 | J NF                              | intuitive | 0.2047733 |
| 3 | NJ F                              | intuitive | 0.2094315 |
| 4 | NJ NF                             | intuitive | 0.2050811 |
| 5 | J F                               | reflexive | 0.6133130 |
| 6 | J NF                              | reflexive | 0.5081431 |
| 7 | NJ F                              | reflexive | 0.5298543 |
| 8 | NJ NF                             | reflexive | 0.4206986 |

## Dot accuracy (concurrent memory task in the intuitive phase 1)

We can check the accuracy of responses to the dot task in the intuitive phase

```
mean(d_long$dot_accuracy)
```

```
[1] 0.9251172
```

```
with(d_long, tapply(dot_accuracy, problem, mean))
```

|  | 1         | 2         | 3         | 4         | 5         | 6         |
|--|-----------|-----------|-----------|-----------|-----------|-----------|
|  | 0.9503013 | 0.9484870 | 0.8775352 | 0.9343614 | 0.9350742 | 0.9049440 |

## Prepare data for modelling

In order to enter the data in the models specified as in the pre-registration, we need some additional steps. We first transform the data such that the responses in the two phases are on distinct columns:

```
d_all <- d_long %>%
  pivot_wider(
    names_from = condition,
    values_from = c(response, accuracy),
    names_sep = "_")
str(d_all)
```

```
tibble [92,598 x 13] (S3: tbl_df/tbl/data.frame)
 $ participant      : int  [1:92598] 1 1 1 1 1 1 2 2 2 2 ...
 $ justification    : chr   [1:92598] "J" "J" "J" "J" ...
 $ feedback         : chr   [1:92598] "NF" "NF" "NF" "NF" ...
 $ problem          : int   [1:92598] 1 2 3 4 5 6 1 2 3 4 ...
 $ familiarity      : chr   [1:92598] "Yes" "No" "No" "No" ...
 $ ext_resources_use : chr   [1:92598] "No" "No" "No" "No" ...
 $ dot_accuracy     : num   [1:92598] 1 1 1 0 1 1 1 1 0 1 ...
 $ Country          : chr   [1:92598] "Canada" "Canada" "Canada" "Canada" ...
 $ culture          : chr   [1:92598] "Individualistic" "Individualistic" "Individualistic" "Individualistic" ...
 $ response_intuitive: num   [1:92598] 2 NA 3 2 2 3 4 1 1 1 ...
 $ response_reflexive: num   [1:92598] 1 1 4 2 2 1 1 1 1 1 ...
 $ accuracy_intuitive: num   [1:92598] 0 NA 0 0 0 0 0 1 1 1 ...
 $ accuracy_reflexive: num   [1:92598] 1 1 0 0 0 1 1 1 1 1 ...
```

## Apply exclusions

The preregistration included further hypothesis-specific exclusion criteria:

For the variable included in Hypothesis 1, we will exclude trials in which participants indicated that they used external resources to solve the problems, reported familiarity with the problem, failed to answer on time, or failed to answer the concurrent working memory load task correctly. For the dependent variable used in Hypotheses 2 and 3, we will exclude trials where the phase 1 response was correct; thus, participants who initially did not commit at least one error in one of the six problems will be excluded. Trials where participants indicate external resource use will also be excluded.

One concern is that excluding based on familiarity response means we are excluding as much as half or more responses for some problems. However, familiarity does seem to influence accuracy, as shown in the table below:

Table 7: Mean accuracy by familiarity CRT problem and phase (here 'intuitive' is phase 1 and 'reflexive' is phase 2)

| justification | feedback | familiarity | intuitive | reflexive |
|---------------|----------|-------------|-----------|-----------|
| J             | F        | No          | 0.20      | 0.65      |
| J             | F        | Yes         | 0.48      | 0.82      |

|    |    |     |      |      |
|----|----|-----|------|------|
| J  | NF | No  | 0.20 | 0.52 |
| J  | NF | Yes | 0.48 | 0.74 |
| NJ | F  | No  | 0.21 | 0.58 |
| NJ | F  | Yes | 0.49 | 0.79 |
| NJ | NF | No  | 0.20 | 0.44 |
| NJ | NF | Yes | 0.50 | 0.71 |

We apply exclusions for the dataset to test H1 as follow

```
d_all_H1 <- d_all %>%
  filter(ext_resources_use=="No",
         familiarity=="No",
         dot_accuracy==1)
```

After exclusions, the mean accuracy values are the following:

Table 8: Mean accuracy by familiarity CRT problem and phase (here 'intuitive' is phase 1 and 'reflexive' is phase 2), after all exclusions.

|    | justification | feedback | intuitive | reflexive |
|----|---------------|----------|-----------|-----------|
| J  |               | F        | 0.20      | 0.65      |
| J  |               | NF       | 0.21      | 0.52      |
| NJ |               | F        | 0.21      | 0.57      |
| NJ |               | NF       | 0.21      | 0.43      |

## Compute dependent variables for specific hypotheses

For Hypothesis 1 ( “less than half of correct answers will arise from error correction during deliberation” ) we restrict analyses to all correct responses, and examine the proportion of correct responses made in phase 2 out of all correct responses:

```
dat_H1 <- d_all_H1 %>%
  filter(accuracy_intuitive == 1 | accuracy_reflexive == 1) %>% # keep only correct, either at phase 1 or 2
  filter(!is.na(accuracy_intuitive)) %>% # excluded missing responses in phase 1
  mutate(correct_phase2 = ifelse(accuracy_intuitive == 0, accuracy_reflexive, 0)) %>%
  select(correct_phase2, participant, problem, justification, feedback, Country, culture)
str(dat_H1)
```

```
tibble [24,590 x 7] (S3: tbl_df/tbl/data.frame)
 $ correct_phase2: num [1:24590] 1 1 0 0 1 0 1 1 1 1 ...
 $ participant   : int [1:24590] 1 2 2 2 2 4 6 6 6 7 ...
 $ problem       : int [1:24590] 6 1 2 4 5 2 3 4 5 1 ...
 $ justification : chr [1:24590] "J" "J" "J" "J" ...
 $ feedback      : chr [1:24590] "NF" "F" "F" "F" ...
 $ Country       : chr [1:24590] "Canada" "Canada" "Canada" "Canada" ...
 $ culture       : chr [1:24590] "Individualistic" "Individualistic" "Individualistic" "Individualistic"
```

*Note that from the dataset for H1 I am excluding missing phase 1 responses. This is because we cannot know whether participant would have responded correctly or not in phase 1 for those problems.*

The remaining hypotheses concern the probability intuitive incorrect answers (errors in phase 1) are corrected in the reflexive phase (phase 2). These hypotheses are not relevant for the pilot since we have only 1 condition. We can still use this data to estimate the probability of correction in the baseline condition.

```
# The other hypotheses
dat_Hother <- d_all %>%
  filter(accuracy_intuitive == 0,          # keep only intuitive errors
         ext_resources_use=="No") %>%      # exclude also external resource use
  mutate(corrected = accuracy_reflexive) %>%
  select(corrected, participant, problem, justification, feedback, Country, culture)
str(dat_Hother)
```

```
tibble [48,693 x 7] (S3: tbl_df/tbl/data.frame)
 $ corrected      : num [1:48693] 1 0 0 0 1 1 1 1 0 0 ...
 $ participant    : int [1:48693] 1 1 1 1 1 2 2 4 4 4 ...
 $ problem        : int [1:48693] 1 3 4 5 6 1 5 1 3 4 ...
 $ justification  : chr [1:48693] "J" "J" "J" "J" ...
 $ feedback       : chr [1:48693] "NF" "NF" "NF" "NF" ...
 $ Country        : chr [1:48693] "Canada" "Canada" "Canada" "Canada" ...
 $ culture        : chr [1:48693] "Individualistic" "Individualistic" "Individualistic" "Individualistic"
```

## Data summaries

### Accuracy

Proportion of correct responses split by item and phase (Note: this is after applying H1 exclusion criteria — thus excluding based on familiarity and use of external resources).

Table 9: Mean accuracy by CRT problem and phase (here 'intuitive' is phase 1 and 'reflexive' is phase 2)

| justification | feedback | problem | intuitive | reflexive |
|---------------|----------|---------|-----------|-----------|
| J             | F        | 1       | 0.20      | 0.75      |
| J             | F        | 2       | 0.43      | 0.85      |
| J             | F        | 3       | 0.10      | 0.53      |
| J             | F        | 4       | 0.23      | 0.75      |
| J             | F        | 5       | 0.08      | 0.47      |
| J             | F        | 6       | 0.22      | 0.62      |
| J             | NF       | 1       | 0.19      | 0.49      |
| J             | NF       | 2       | 0.42      | 0.75      |
| J             | NF       | 3       | 0.10      | 0.49      |
| J             | NF       | 4       | 0.25      | 0.57      |
| J             | NF       | 5       | 0.09      | 0.35      |
| J             | NF       | 6       | 0.22      | 0.48      |
| NJ            | F        | 1       | 0.21      | 0.75      |
| NJ            | F        | 2       | 0.40      | 0.81      |
| NJ            | F        | 3       | 0.11      | 0.38      |
| NJ            | F        | 4       | 0.27      | 0.71      |
| NJ            | F        | 5       | 0.09      | 0.35      |
| NJ            | F        | 6       | 0.23      | 0.55      |
| NJ            | NF       | 1       | 0.18      | 0.44      |
| NJ            | NF       | 2       | 0.41      | 0.67      |
| NJ            | NF       | 3       | 0.11      | 0.33      |
| NJ            | NF       | 4       | 0.25      | 0.50      |
| NJ            | NF       | 5       | 0.10      | 0.29      |
| NJ            | NF       | 6       | 0.23      | 0.39      |

## Missing responses

The number of missing responses (trials in which participants did not respond in time)

Table 10: Proportion and number of missing responses in phase 1 (intuitive)

| problem | prop_missing | N_missing |
|---------|--------------|-----------|
| 1       | 0.04         | 542       |
| 2       | 0.14         | 2179      |
| 3       | 0.28         | 4351      |
| 4       | 0.09         | 1355      |
| 5       | 0.16         | 2543      |
| 6       | 0.27         | 4243      |

## Responses correct in phase 1 and turned into errors in phase 2

Using the data transformed as in the object `d_all` above we can check how frequently participants made a correct response in phase 1, and changed it into an error in phase 2. The table below shows how many times this occurred for each CRT problem:

```
correct2error_table <- d_all %>%  
  filter(accuracy_intuitive == 1 | accuracy_reflexive == 1) %>%  
  select(accuracy_intuitive, accuracy_reflexive, problem) %>%  
  filter(accuracy_reflexive == 0) %>%  
  group_by(problem) %>%  
  summarise(N = sum(accuracy_intuitive))  
  
print(correct2error_table)
```

```
# A tibble: 6 x 2  
  problem      N  
  <int> <dbl>  
1       1  110  
2       2  337  
3       3  275  
4       4  549  
5       5  353  
6       6  501
```

Overall across all problems this occurred 2125 times.

## Confidence calibration

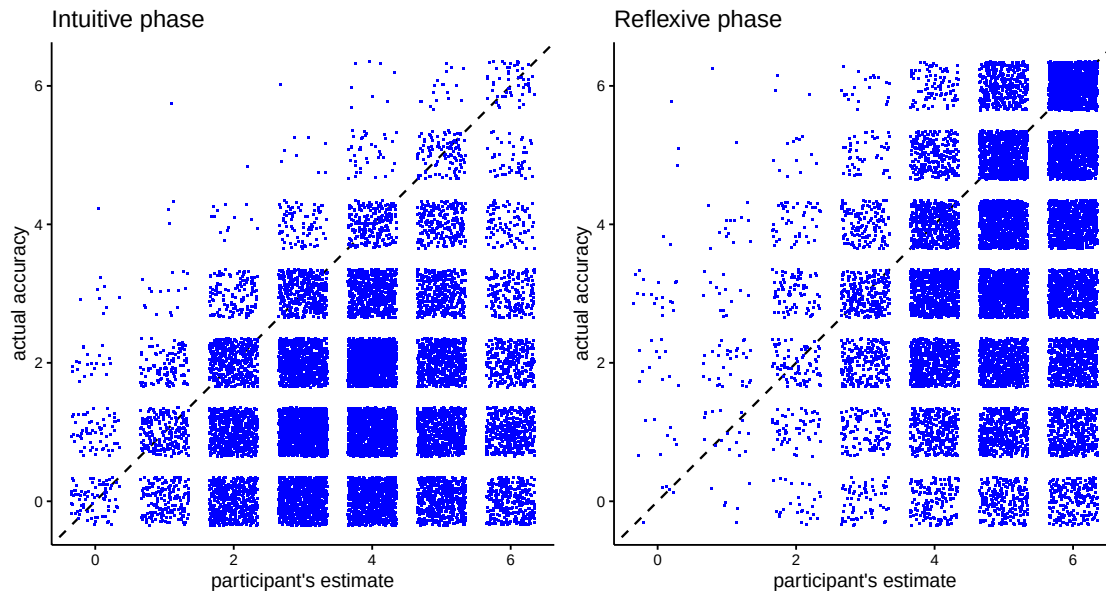


Figure 1: Number of correct answer (vertical coordinates) plotted as a function of the answer to the calibration responses (e.g. the participant's estimates of their own accuracy; horizontal coordinates). Some jitter added for visibility. Both plots demonstrates substantial overconfidence, since in both cases most points lie below the identity line. For this plot we only excluded problems where participants declared using external resources.

### Confidence error and confidence bias

For this type of question, the squared error provide a proper scoring rule assuming participant's expected number correct (squared error is uniquely minimised, in expectation, by reporting the subjective mean).

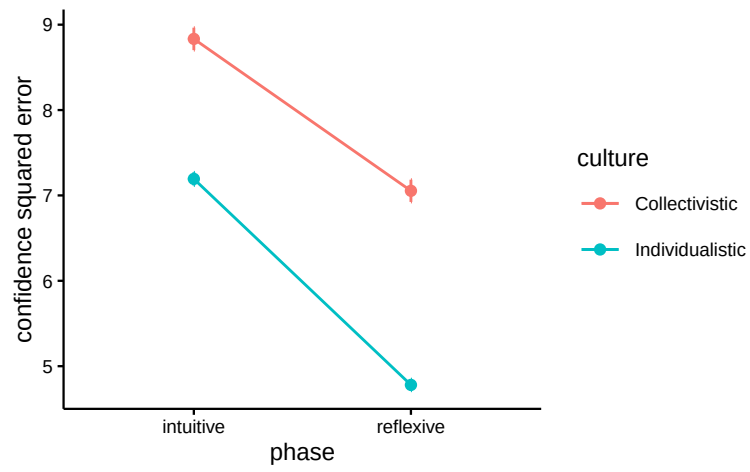


Figure 2: Confidence error, quantified as the squared error between the predicted and actual number of correct problems, split by phase and cultural category. Error bars are standard error of the mean.

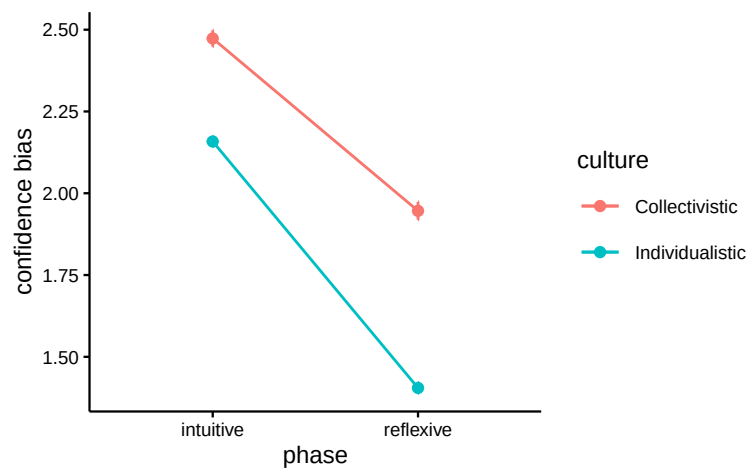


Figure 3: Confidence bias, quantified as the signed difference the predicted and actual number of correct problems (so bias >0 indicate over-confidence), split by phase and cultural category. Error bars are standard error of the mean.

## Confidence squared-error maps

This is purely explorative — I tried to do some maps of the confidence calibration question.

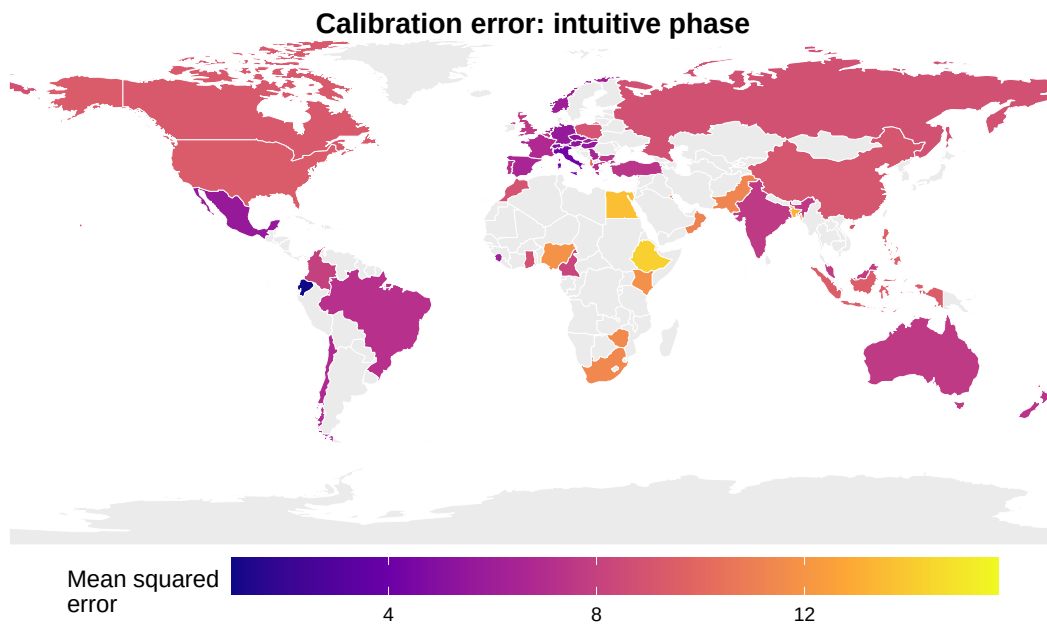


Figure 4: Confidence squared error map (intuitive phase).

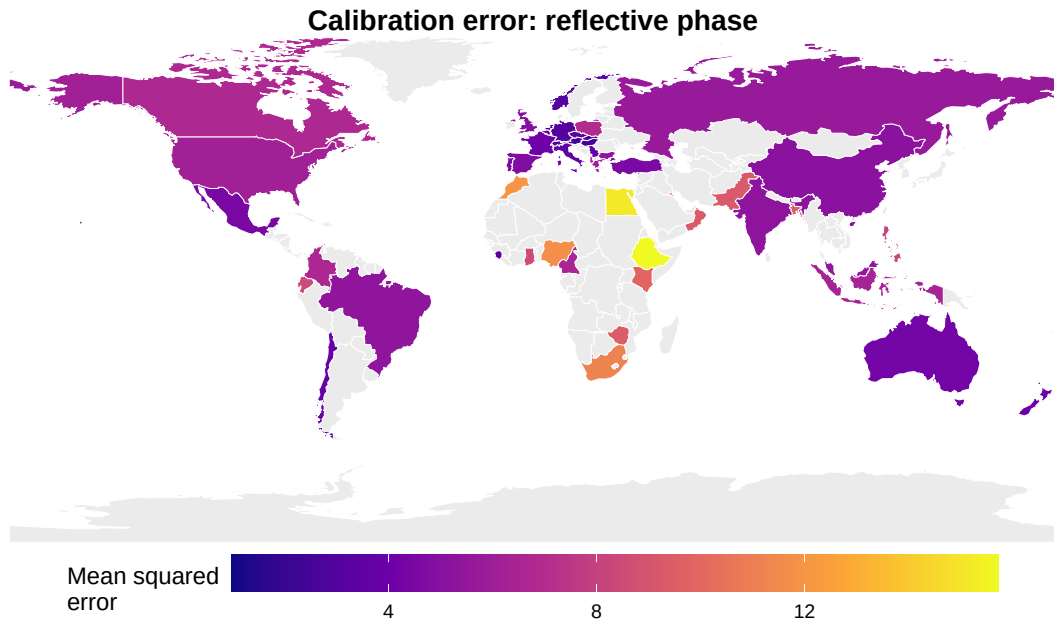


Figure 5: Confidence squared error map (reflexive phase).

### Confidence bias maps

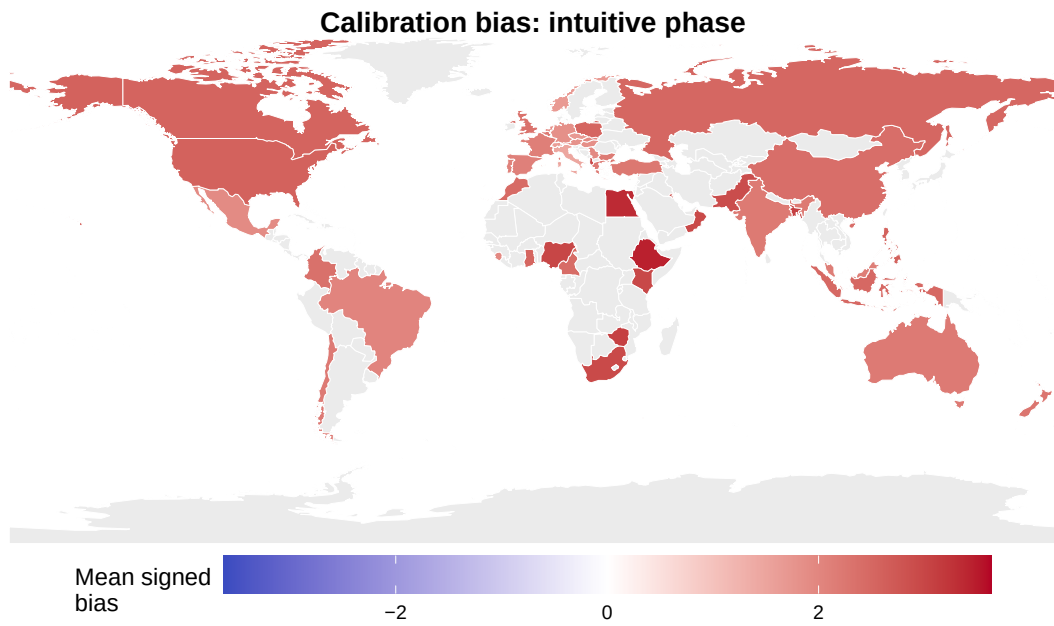


Figure 6: Confidence bias map (intuitive phase).

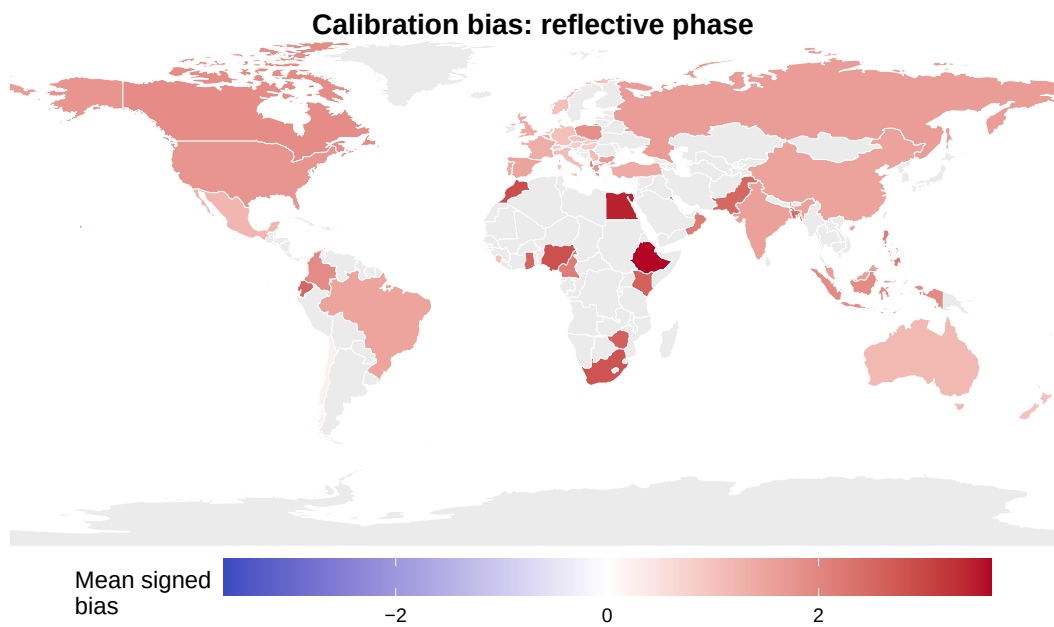


Figure 7: Confidence bias map (reflexive phase).

## Analyses

Recode justification and feedback as dummy variables

```
dat_H1$justification <- ifelse(dat_H1$justification=="J", 1, 0)
dat_H1$feedback <- ifelse(dat_H1$feedback=="F", 1, 0)

dat_Hother$justification <- ifelse(dat_Hother$justification=="J", 1, 0)
dat_Hother$feedback <- ifelse(dat_Hother$feedback=="F", 1, 0)
```

### H1

A plot of the variable for H1, that is the probability that a correct decision is made in phase 2 rather than in phase 1, already suggests that in general this probability is not below 0.5

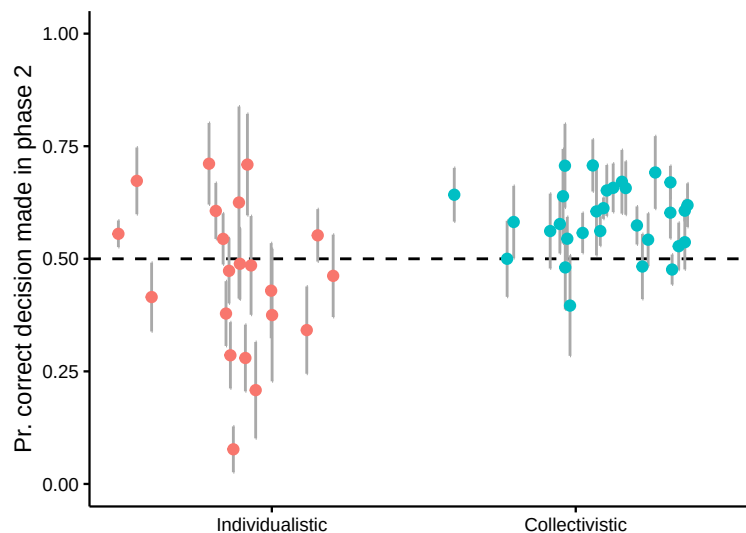


Figure 8: Probability that a correct decision is made in phase 2 rather than in phase 1 (H1 variable), split by country and culture. Error bars are standard errors between participants in each country.

Fit models for H1

```
mH1_all <- glmer(correct_phase2 ~ feedback * justification +
  (feedback * justification||Country),
  family = binomial("logit"), data = dat_H1,
  control = glmerControl(optimizer = "bobyqa",
    optCtrl = list(maxfun = 1e5)))

summary(mH1_all)
```

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [glmerMod]  
Family: binomial ( logit )  
Formula:

```
correct_phase2 ~ feedback * justification + (feedback * justification ||
  Country)
Data: dat_H1
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 31542.3 | 31607.2 | -15763.2 | 31526.3  | 24582    |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.0704 | -1.2070 | 0.6378 | 0.7555 | 1.6492 |

Random effects:

| Groups    | Name                   | Variance  | Std.Dev.  |
|-----------|------------------------|-----------|-----------|
| Country   | (Intercept)            | 2.032e-01 | 4.507e-01 |
| Country.1 | feedback               | 4.988e-10 | 2.233e-05 |
| Country.2 | justification          | 4.144e-17 | 6.437e-09 |
| Country.3 | feedback:justification | 6.957e-13 | 8.341e-07 |

Number of obs: 24590, groups: Country, 53

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )     |
|------------------------|----------|------------|---------|--------------|
| (Intercept)            | 0.13962  | 0.07016    | 1.990   | 0.0466 *     |
| feedback               | 0.34753  | 0.03831    | 9.072   | < 2e-16 ***  |
| justification          | 0.24985  | 0.03946    | 6.332   | 2.42e-10 *** |
| feedback:justification | -0.03926 | 0.05413    | -0.725  | 0.4683       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|  | (Intr) | fedbck | jstfct |
|--|--------|--------|--------|
|--|--------|--------|--------|

|          |        |  |  |
|----------|--------|--|--|
| feedback | -0.297 |  |  |
|----------|--------|--|--|

|             |        |       |  |
|-------------|--------|-------|--|
| justificatn | -0.286 | 0.522 |  |
|-------------|--------|-------|--|

|             |       |        |        |
|-------------|-------|--------|--------|
| fdbck:jstfc | 0.209 | -0.707 | -0.728 |
|-------------|-------|--------|--------|

optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

```
# fit country_group-level models for individualistic and collectivistic
mH1_i <- glmer(correct_phase2 ~ feedback * justification +
  (feedback * justification||Country),
  family = binomial("logit"),
  data = dat_H1[dat_H1$culture == "Individualistic", ],
  control = glmerControl(optimizer = "bobyqa",
    optCtrl = list(maxfun = 1e5)))

summary(mH1_i)
```

Generalized linear mixed model fit by maximum likelihood (Laplace  
Approximation) [glmerMod]

Family: binomial ( logit )

Formula:

```
correct_phase2 ~ feedback * justification + (feedback * justification ||
  Country)
```

```
Data: dat_H1[dat_H1$culture == "Individualistic", ]
```

```
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 23594.1 | 23656.8 | -11789.1 | 23578.1  | 18617    |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.0374 | -1.2370 | 0.6316 | 0.7284 | 1.0827 |

Random effects:

| Groups    | Name                   | Variance  | Std.Dev.  |
|-----------|------------------------|-----------|-----------|
| Country   | (Intercept)            | 5.515e-02 | 2.348e-01 |
| Country.1 | feedback               | 1.893e-03 | 4.350e-02 |
| Country.2 | justification          | 3.696e-14 | 1.922e-07 |
| Country.3 | feedback:justification | 0.000e+00 | 0.000e+00 |

Number of obs: 18625, groups: Country, 31

Fixed effects:

|                        | Estimate  | Std. Error | z value | Pr(> z )     |
|------------------------|-----------|------------|---------|--------------|
| (Intercept)            | 0.343689  | 0.056019   | 6.135   | 8.51e-10 *** |
| feedback               | 0.341169  | 0.048350   | 7.056   | 1.71e-12 *** |
| justification          | 0.243230  | 0.045456   | 5.351   | 8.75e-08 *** |
| feedback:justification | -0.008226 | 0.062592   | -0.131  | 0.895        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    | -0.449 |        |        |
| justificatn | -0.410 | 0.478  |        |
| fdbck:jstfc | 0.294  | -0.642 | -0.725 |

optimizer (bobyqa) convergence code: 0 (OK)  
boundary (singular) fit: see help('isSingular')

```
mH1_c <- glmer(correct_phase2 ~ feedback * justification +  
  (feedback * justification||Country),  
  family = binomial("logit"),  
  data = dat_H1[dat_H1$culture == "Collectivistic", ],  
  control = glmerControl(optimizer = "bobyqa",  
    optCtrl = list(maxfun = 1e5)))  
  
summary(mH1_c)
```

Generalized linear mixed model fit by maximum likelihood (Laplace  
Approximation) [glmerMod]

Family: binomial ( logit )

Formula:

correct\_phase2 ~ feedback \* justification + (feedback \* justification ||  
Country)

Data: dat\_H1[dat\_H1\$culture == "Collectivistic", ]

Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))

| AIC    | BIC    | logLik  | deviance | df.resid |
|--------|--------|---------|----------|----------|
| 7933.4 | 7986.9 | -3958.7 | 7917.4   | 5957     |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -1.7087 | -1.1172 | 0.6993 | 0.8188 | 1.7073 |

Random effects:

| Groups    | Name                   | Variance | Std.Dev. |
|-----------|------------------------|----------|----------|
| Country   | (Intercept)            | 0.2872   | 0.5359   |
| Country.1 | feedback               | 0.0000   | 0.0000   |
| Country.2 | justification          | 0.0000   | 0.0000   |
| Country.3 | feedback:justification | 0.0000   | 0.0000   |

Number of obs: 5965, groups: Country, 22

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )     |
|------------------------|----------|------------|---------|--------------|
| (Intercept)            | -0.16076 | 0.13205    | -1.217  | 0.223428     |
| feedback               | 0.35670  | 0.07672    | 4.650   | 3.33e-06 *** |
| justification          | 0.26644  | 0.07942    | 3.355   | 0.000794 *** |
| feedback:justification | -0.12949 | 0.10798    | -1.199  | 0.230475     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|  | (Intr) | fedbck | jstfct |
|--|--------|--------|--------|
|--|--------|--------|--------|

|          |        |  |  |
|----------|--------|--|--|
| feedback | -0.329 |  |  |
|----------|--------|--|--|

|             |        |       |  |
|-------------|--------|-------|--|
| justificatn | -0.313 | 0.536 |  |
|-------------|--------|-------|--|

|             |       |        |        |
|-------------|-------|--------|--------|
| fdbck:jstfc | 0.232 | -0.710 | -0.733 |
|-------------|-------|--------|--------|

optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

## Note

The model formula was preregistered as `correct_phase2 ~ feedback * justification + (feedback * justification || Country)`, however this gives a singular model warning. Convergence is OK, so fixed-effects estimates should be usable and are essentially unchanged compared to a simplified model (tested with the below direct comparison), so I am sticking to the pre-registered version.

```
m_full <- glmer(
  correct_phase2 ~ feedback * justification +
    (feedback * justification || Country),
  family = binomial,
  data = dat_H1
)

m_simple <- glmer(
  correct_phase2 ~ feedback * justification +
    (feedback || Country),
  family = binomial,
  data = dat_H1
)

cbind(
  full = fixef(m_full),
  simple = fixef(m_simple)
)
```

|                        | full        | simple      |
|------------------------|-------------|-------------|
| (Intercept)            | 0.13963470  | 0.13962229  |
| feedback               | 0.34752667  | 0.34753149  |
| justification          | 0.24985294  | 0.24985523  |
| feedback:justification | -0.03925052 | -0.03925502 |

```
cbind(
  full = sqrt(diag(vcov(m_full))),
  simple = sqrt(diag(vcov(m_simple)))
)
```

|                        | full       | simple     |
|------------------------|------------|------------|
| (Intercept)            | 0.07021097 | 0.07021526 |
| feedback               | 0.03832611 | 0.03833518 |
| justification          | 0.03947797 | 0.03948385 |
| feedback:justification | 0.05416177 | 0.05417391 |

The first hypothesis is about the proportion of correct responses made in the reflective phase. We predicted that this would be lower than 50% in the baseline condition. It can be computed by transforming the intercept parameter from log-odds to probability:

```
exp(fixef(mH1_all)['(Intercept)'])/(1+exp(fixef(mH1_all)['(Intercept)'])))
```

```
(Intercept)
0.5348494
```

The hypothesis testing code, taken from the power analysis script and implementing one-tailed tests was:

```
alpha <- 0.05
# note: the tests on separate country_groups are independent one another
# (different countries, participants, labs, etc.), so the expected
# false positive rate of the composite test, run on the 2 country_group
# separately and using the conventional alpha = 0.05 for each test,
# would be 0.05^2 = 0.0025 (the hypothesis supported if and only if
# all 2 independent tests are significant). We don't also apply correction
# for multiple comparisons otherwise it would become too conservative.

# -----
# H1

# One-tailed test for H1a
# Testing if the intercept is significantly less than 0
b_H1a <- fixef(mH1_all)["(Intercept)"]
SE_H1a <- sqrt(diag(vcov(mH1_all))["(Intercept)"])
z_H1a <- b_H1a / SE_H1a
p_value_H1a <- pnorm(z_H1a) # left-tailed test
H1a <- ifelse(p_value_H1a < alpha, 1, 0)

# One-tailed tests for H1b
models <- list(mH1_i, mH1_c)
names <- c("Individualistic", "Collectivistic")
H1b_pvals <- numeric(length(models))
H1b_results <- numeric(length(models))

for (i in 1:length(models)) {
  b <- fixef(models[[i]])["(Intercept)"]
  SE <- sqrt(diag(vcov(models[[i]]))["(Intercept)"])
  z <- b / SE
  p_value <- pnorm(z) # left-tailed test
  H1b_pvals[i] <- p_value
  H1b_results[i] <- ifelse(p_value < alpha, 1, 0)
}

names(H1b_pvals) <- names
names(H1b_results) <- names

# complete test
H1b <- ifelse(all(H1b_results == 1), 1, 0)
```

## H2

The models for the remaining hypotheses can be estimated as

```
# fit overall model
m_all <- glmer(corrected ~ feedback * justification +
  (feedback * justification||Country),
  family = binomial("logit"),
  data = dat_Hother,
  control = glmerControl(optimizer = "bobyqa",
    optCtrl = list(maxfun = 1e5)))

summary(m_all)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: corrected ~ feedback * justification + (feedback * justification ||
Country)
Data: dat_Hother
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 64931.6 | 65002.0 | -32457.8 | 64915.6  | 48685    |

Scaled residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -2.2513 | -0.9391 | -0.5367 | 0.9515 | 2.8098 |

Random effects:

| Groups    | Name                   | Variance | Std.Dev. |
|-----------|------------------------|----------|----------|
| Country   | (Intercept)            | 0.224043 | 0.47333  |
| Country.1 | feedback               | 0.001666 | 0.04082  |
| Country.2 | justification          | 0.018484 | 0.13596  |
| Country.3 | feedback:justification | 0.022398 | 0.14966  |

Number of obs: 48693, groups: Country, 53

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )     |
|------------------------|----------|------------|---------|--------------|
| (Intercept)            | -0.56667 | 0.06927    | -8.180  | 2.83e-16 *** |
| feedback               | 0.59802  | 0.02753    | 21.720  | < 2e-16 ***  |
| justification          | 0.38150  | 0.03573    | 10.676  | < 2e-16 ***  |
| feedback:justification | -0.04113 | 0.04634    | -0.888  | 0.375        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    |        | -0.195 |        |
| justificatn | -0.177 |        | 0.366  |
| fdbck:jstfc | 0.102  | -0.552 | -0.483 |

```
# fit country_group-level models for weird and non-weird
m_i <- glmer(corrected ~ feedback * justification +
  (feedback * justification||Country),
  family = binomial("logit"),
  data = dat_Hother[dat_Hother$culture == "Individualistic", ],
  control = glmerControl(optimizer = "bobyqa",
    optCtrl = list(maxfun = 1e5)))

summary(m_i)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: corrected ~ feedback * justification + (feedback * justification ||
Country)
Data: dat_Hother[dat_Hother$culture == "Individualistic", ]
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 46331.1 | 46398.8 | -23157.6 | 46315.1  | 34607    |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.3763 | -0.9639 | 0.5387 | 0.9237 | 1.7404 |

Random effects:

| Groups    | Name                   | Variance  | Std.Dev.  |
|-----------|------------------------|-----------|-----------|
| Country   | (Intercept)            | 1.180e-01 | 3.436e-01 |
| Country.1 | feedback               | 0.000e+00 | 0.000e+00 |
| Country.2 | justification          | 1.432e-02 | 1.197e-01 |
| Country.3 | feedback:justification | 9.008e-10 | 3.001e-05 |

Number of obs: 34615, groups: Country, 31

Fixed effects:

|                        | Estimate  | Std. Error | z value | Pr(> z )     |
|------------------------|-----------|------------|---------|--------------|
| (Intercept)            | -0.380863 | 0.066947   | -5.689  | 1.28e-08 *** |
| feedback               | 0.602197  | 0.030782   | 19.564  | < 2e-16 ***  |
| justification          | 0.403213  | 0.040797   | 9.883   | < 2e-16 ***  |
| feedback:justification | 0.008418  | 0.044177   | 0.191   | 0.849        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    | -0.233 |        |        |
| justificatn | -0.214 | 0.383  |        |
| fdbck:jstfc | 0.160  | -0.696 | -0.540 |

optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

```
m_c <- glmer(corrected ~ feedback * justification +
  (feedback * justification||Country),
  family = binomial("logit"),
```

```
data = dat_Hother[dat_Hother$culture == "Collectivistic", ],
control = glmerControl(optimizer = "bobyqa",
  optCtrl = list(maxfun = 1e5)))

summary(m_c)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: corrected ~ feedback * justification + (feedback * justification ||
Country)
Data: dat_Hother[dat_Hother$culture == "Collectivistic", ]
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik  | deviance | df.resid |
|---------|---------|---------|----------|----------|
| 18589.4 | 18649.8 | -9286.7 | 18573.4  | 14070    |

Scaled residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.5734 | -0.8647 | -0.5980 | 1.0418 | 2.8459 |

Random effects:

| Groups    | Name                   | Variance | Std.Dev. |
|-----------|------------------------|----------|----------|
| Country   | (Intercept)            | 0.235623 | 0.48541  |
| Country.1 | feedback               | 0.004018 | 0.06339  |
| Country.2 | justification          | 0.024461 | 0.15640  |
| Country.3 | feedback:justification | 0.046696 | 0.21609  |

Number of obs: 14078, groups: Country, 22

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )     |
|------------------------|----------|------------|---------|--------------|
| (Intercept)            | -0.83668 | 0.11418    | -7.328  | 2.34e-13 *** |
| feedback               | 0.60046  | 0.05721    | 10.495  | < 2e-16 ***  |
| justification          | 0.35362  | 0.06746    | 5.242   | 1.59e-07 *** |
| feedback:justification | -0.15128 | 0.09346    | -1.619  | 0.106        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    | -0.253 |        |        |
| justificatn | -0.224 | 0.402  |        |
| fdbck:jstfc | 0.140  | -0.552 | -0.494 |

Test pre-registered hypotheses

```
alpha <- 0.05

# -----
# H2

# feedback increases correction rate
```

```

# One-tailed test for H2a
# Testing if the coefficient for 'feedback' is significantly greater than 0
b_H2a <- fixef(m_all)["feedback"]
SE_H2a <- sqrt(diag(vcov(m_all))["feedback"])
z_H2a <- b_H2a / SE_H2a
p_value_H2a <- 1 - pnorm(z_H2a) # right-tailed test
H2a <- ifelse(p_value_H2a < alpha, 1, 0)

# One-tailed tests for H2b
models <- list(m_i, m_c)
names <- c("Individualistic", "Collectivistic")
H2b_pvals <- numeric(length(models))
H2b_results <- numeric(length(models))

for (i in 1:length(models)) {
  b <- fixef(models[[i]])["feedback"]
  SE <- sqrt(diag(vcov(models[[i]]))["feedback"])
  z <- b / SE
  p_value <- 1 - pnorm(z) # right-tailed test
  H2b_pvals[i] <- p_value
  H2b_results[i] <- ifelse(p_value < alpha, 1, 0)
}

names(H2b_pvals) <- names
names(H2b_results) <- names

```

### H3

```
# -----  
# H3  
  
# justification increases correction rate  
  
# One-tailed test for H3a  
# Testing if the coefficient for 'justification' is significantly greater than 0  
b_H3a <- fixef(m_all)["justification"]  
SE_H3a <- sqrt(diag(vcov(m_all))["justification"])  
z_H3a <- b_H3a / SE_H3a  
p_value_H3a <- 1 - pnorm(z_H3a) # right-tailed test  
H3a <- ifelse(p_value_H3a < alpha, 1, 0)  
  
# One-tailed tests for H3b  
models <- list(m_i, m_c)  
names <- c("Individualistic", "Collectivistic")  
H3b_pvals <- numeric(length(models))  
H3b_results <- numeric(length(models))  
  
for (i in 1:length(models)) {  
  b <- fixef(models[[i]])["justification"]  
  SE <- sqrt(diag(vcov(models[[i]]))["justification"])  
  z <- b / SE  
  p_value <- 1 - pnorm(z) # right-tailed test  
  H3b_pvals[i] <- p_value  
  H3b_results[i] <- ifelse(p_value < alpha, 1, 0)  
}  
  
names(H3b_pvals) <- names  
names(H3b_results) <- names
```

## Confirmatory hypothesis testing summary (pre-registered analyses)

Table 11: Summary of preregistered hypothesis tests after including by-problem random intercepts

| Hypothesis | Prediction                                                                             | Supported | p-value(s)                                        |
|------------|----------------------------------------------------------------------------------------|-----------|---------------------------------------------------|
| H1a        | Less than half of correct answers will arise from error correction during deliberation | No        | .977                                              |
| H1b        | Baseline performance is below 50% in both cultural groups                              | No        | Individualistic = 1.000; Collectivistic = .112    |
| H2a        | Feedback increases correction overall                                                  | Yes       | < .001                                            |
| H2b        | Feedback increases correction in both cultural groups                                  | Yes       | Individualistic = < .001; Collectivistic = < .001 |
| H3a        | Justification increases correction overall                                             | Yes       | < .001                                            |
| H3b        | Justification increases correction in both cultural groups                             | Yes       | Individualistic = < .001; Collectivistic = < .001 |

## Robustness & sensitivity checks

### By-problem random intercepts

As problems differed substantially in accuracy etc. it makes sense to add problem-specific random intercepts

#### H1

Fit models for H1

```
mH1_all <- glmer(correct_phase2 ~ feedback * justification +  
  (feedback * justification||Country)+ (1|problem),  
  family = binomial("logit"), data = dat_H1,  
  control = glmerControl(optimizer = "bobyqa",  
    optCtrl = list(maxfun = 1e5)))  
  
summary(mH1_all )
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace  
Approximation) [glmerMod]  
Family: binomial ( logit )  
Formula:  
correct_phase2 ~ feedback * justification + (feedback * justification ||  
Country) + (1 | problem)  
Data: dat_H1  
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 30471.6 | 30544.6 | -15226.8 | 30453.6  | 24581    |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.9853 | -1.0537 | 0.5404 | 0.7370 | 2.3630 |

Random effects:

| Groups    | Name                   | Variance | Std.Dev. |
|-----------|------------------------|----------|----------|
| Country   | (Intercept)            | 0.220492 | 0.46957  |
| Country.1 | feedback               | 0.006141 | 0.07837  |
| Country.2 | justification          | 0.000000 | 0.00000  |
| Country.3 | feedback:justification | 0.002225 | 0.04717  |
| problem   | (Intercept)            | 0.220268 | 0.46933  |

Number of obs: 24590, groups: Country, 53; problem, 6

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )     |
|------------------------|----------|------------|---------|--------------|
| (Intercept)            | 0.23460  | 0.20436    | 1.148   | 0.251        |
| feedback               | 0.35109  | 0.04338    | 8.093   | 5.84e-16 *** |
| justification          | 0.24297  | 0.04046    | 6.006   | 1.90e-09 *** |
| feedback:justification | -0.05016 | 0.05633    | -0.891  | 0.373        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```

Correlation of Fixed Effects:
      (Intr) fedbck jstfct
feedback      -0.107
justificatn -0.100  0.471
fdbck:jstfc  0.070 -0.626 -0.717
optimizer (bobyqa) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

```

```

# fit country_group-level models for individualistic and collectivistic
mH1_i <- glmer(correct_phase2 ~ feedback * justification +
  (feedback * justification||Country)+ (1|problem),
  family = binomial("logit"),
  data = dat_H1[dat_H1$culture == "Individualistic", ],
  control = glmerControl(optimizer = "bobyqa",
    optCtrl = list(maxfun = 1e5)))

summary(mH1_i)

```

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula:
correct_phase2 ~ feedback * justification + (feedback * justification ||
Country) + (1 | problem)
Data: dat_H1[dat_H1$culture == "Individualistic", ]
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))

```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 22676.4 | 22746.9 | -11329.2 | 22658.4  | 18616    |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -3.0407 | -1.0719 | 0.5202 | 0.7236 | 1.5416 |

Random effects:

| Groups    | Name                   | Variance  | Std.Dev.  |
|-----------|------------------------|-----------|-----------|
| Country   | (Intercept)            | 6.673e-02 | 2.583e-01 |
| Country.1 | feedback               | 1.018e-02 | 1.009e-01 |
| Country.2 | justification          | 1.895e-13 | 4.354e-07 |
| Country.3 | feedback:justification | 2.273e-03 | 4.768e-02 |
| problem   | (Intercept)            | 2.567e-01 | 5.066e-01 |

Number of obs: 18625, groups: Country, 31; problem, 6

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )     |
|------------------------|----------|------------|---------|--------------|
| (Intercept)            | 0.46002  | 0.21514    | 2.138   | 0.0325 *     |
| feedback               | 0.33833  | 0.05220    | 6.481   | 9.10e-11 *** |
| justification          | 0.23232  | 0.04679    | 4.966   | 6.85e-07 *** |
| feedback:justification | -0.01644 | 0.06560    | -0.251  | 0.8021       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```

Correlation of Fixed Effects:
      (Intr) fedbck jstfct

```

```

feedback      -0.112
justificatn   -0.109  0.451
fdbck:jstfc   0.076 -0.605 -0.713
optimizer (bobyqa) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

```

```

mH1_c <- glmer(correct_phase2 ~ feedback * justification +
  (feedback * justification||Country)+ (1|problem),
  family = binomial("logit"),
  data = dat_H1[dat_H1$culture == "Collectivistic", ],
  control = glmerControl(optimizer = "bobyqa",
    optCtrl = list(maxfun = 1e5)))

```

```
summary(mH1_c)
```

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [glmerMod]

Family: binomial ( logit )

Formula:

```
correct_phase2 ~ feedback * justification + (feedback * justification ||
  Country) + (1 | problem)
```

Data: dat\_H1[dat\_H1\$culture == "Collectivistic", ]

Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))

| AIC    | BIC    | logLik  | deviance | df.resid |
|--------|--------|---------|----------|----------|
| 7773.2 | 7833.4 | -3877.6 | 7755.2   | 5956     |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.1071 | -1.0037 | 0.6054 | 0.7915 | 2.3848 |

Random effects:

| Groups    | Name                   | Variance  | Std.Dev.  |
|-----------|------------------------|-----------|-----------|
| Country   | (Intercept)            | 3.075e-01 | 5.546e-01 |
| Country.1 | feedback               | 0.000e+00 | 0.000e+00 |
| Country.2 | justification          | 0.000e+00 | 0.000e+00 |
| Country.3 | feedback:justification | 1.416e-12 | 1.190e-06 |
| problem   | (Intercept)            | 1.419e-01 | 3.767e-01 |

Number of obs: 5965, groups: Country, 22; problem, 6

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )     |
|------------------------|----------|------------|---------|--------------|
| (Intercept)            | -0.10527 | 0.20543    | -0.512  | 0.608325     |
| feedback               | 0.36774  | 0.07796    | 4.717   | 2.39e-06 *** |
| justification          | 0.26573  | 0.08071    | 3.293   | 0.000993 *** |
| feedback:justification | -0.13592 | 0.10967    | -1.239  | 0.215219     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    | -0.213 |        |        |
| justificatn | -0.204 | 0.535  |        |
| fdbck:jstfc | 0.152  | -0.709 | -0.734 |

```
optimizer (bobyqa) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

```
alpha <- 0.05
# note: the tests on separate country_groups are independent one another
# (different countries, participants, labs, etc.), so the expected
# false positive rate of the composite test, run on the 2 country_group
# separately and using the conventional alpha = 0.05 for each test,
# would be  $0.05^2 = 0.0025$  (the hypothesis supported if and only if
# all 2 independent tests are significant). We don't also apply correction
# for multiple comparisons otherwise it would become too conservative.

# -----
# H1

# One-tailed test for H1a
# Testing if the intercept is significantly less than 0
b_H1a <- fixef(mH1_all)["(Intercept)"]
SE_H1a <- sqrt(diag(vcov(mH1_all))["(Intercept)"])
z_H1a <- b_H1a / SE_H1a
p_value_H1a <- pnorm(z_H1a) # left-tailed test
H1a <- ifelse(p_value_H1a < alpha, 1, 0)

# One-tailed tests for H1b
models <- list(mH1_i, mH1_c)
names <- c("Individualistic", "Collectivistic")
H1b_pvals <- numeric(length(models))
H1b_results <- numeric(length(models))

for (i in 1:length(models)) {
  b <- fixef(models[[i]])["(Intercept)"]
  SE <- sqrt(diag(vcov(models[[i]]))["(Intercept)"])
  z <- b / SE
  p_value <- pnorm(z) # left-tailed test
  H1b_pvals[i] <- p_value
  H1b_results[i] <- ifelse(p_value < alpha, 1, 0)
}

names(H1b_pvals) <- names
names(H1b_results) <- names
```

## H2

```
# fit overall model
m_all <- glmer(corrected ~ feedback * justification +
  (feedback * justification||Country)+ (1|problem),
  family = binomial("logit"),
  data = dat_Hother,
  control = glmerControl(optimizer = "bobyqa",
    optCtrl = list(maxfun = 1e5)))

summary(m_all)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: corrected ~ feedback * justification + (feedback * justification ||
  Country) + (1 | problem)
Data: dat_Hother
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 62221.0 | 62300.1 | -31101.5 | 62203.0  | 48684    |

Scaled residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -3.3763 | -0.8781 | -0.4479 | 0.8819 | 3.3383 |

Random effects:

| Groups    | Name                   | Variance | Std.Dev. |
|-----------|------------------------|----------|----------|
| Country   | (Intercept)            | 0.259202 | 0.50912  |
| Country.1 | feedback               | 0.003227 | 0.05681  |
| Country.2 | justification          | 0.023606 | 0.15364  |
| Country.3 | feedback:justification | 0.027499 | 0.16583  |
| problem   | (Intercept)            | 0.269691 | 0.51932  |

Number of obs: 48693, groups: Country, 53; problem, 6

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )   |
|------------------------|----------|------------|---------|------------|
| (Intercept)            | -0.54248 | 0.22445    | -2.417  | 0.0157 *   |
| feedback               | 0.63729  | 0.02941    | 21.671  | <2e-16 *** |
| justification          | 0.40972  | 0.03818    | 10.732  | <2e-16 *** |
| feedback:justification | -0.04910 | 0.04883    | -1.006  | 0.3146     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    | -0.062 |        |        |
| justificatn | -0.056 | 0.348  |        |
| fdbck:jstfc | 0.033  | -0.533 | -0.466 |

```
# fit country_group-level models for weird and non-weird
m_i <- glmer(corrected ~ feedback * justification +
```

```

    (feedback * justification||Country)+ (1|problem),
    family = binomial("logit"),
    data = dat_Hother[dat_Hother$culture == "Individualistic", ],
    control = glmerControl(optimizer = "bobyqa",
        optCtrl = list(maxfun = 1e5)))

summary(m_i)

```

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: corrected ~ feedback * justification + (feedback * justification ||
Country) + (1 | problem)
Data: dat_Hother[dat_Hother$culture == "Individualistic", ]
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))

```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 44183.2 | 44259.3 | -22082.6 | 44165.2  | 34606    |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -3.7000 | -0.8911 | 0.4320 | 0.8535 | 2.4609 |

Random effects:

| Groups    | Name                   | Variance  | Std.Dev.  |
|-----------|------------------------|-----------|-----------|
| Country   | (Intercept)            | 1.458e-01 | 3.818e-01 |
| Country.1 | feedback               | 2.023e-14 | 1.422e-07 |
| Country.2 | justification          | 1.839e-02 | 1.356e-01 |
| Country.3 | feedback:justification | 2.576e-13 | 5.075e-07 |
| problem   | (Intercept)            | 3.049e-01 | 5.522e-01 |

Number of obs: 34615, groups: Country, 31; problem, 6

Fixed effects:

|                        | Estimate  | Std. Error | z value | Pr(> z )   |
|------------------------|-----------|------------|---------|------------|
| (Intercept)            | -0.344653 | 0.236347   | -1.458  | 0.145      |
| feedback               | 0.647615  | 0.031872   | 20.319  | <2e-16 *** |
| justification          | 0.435376  | 0.043599   | 9.986   | <2e-16 *** |
| feedback:justification | 0.002235  | 0.045641   | 0.049   | 0.961      |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    | -0.068 |        |        |
| justificatn | -0.062 | 0.372  |        |
| fdbck:jstfc | 0.047  | -0.696 | -0.523 |

optimizer (bobyqa) convergence code: 0 (OK)  
boundary (singular) fit: see help('isSingular')

```

m_c <- glmer(corrected ~ feedback * justification +
    (feedback * justification||Country)+ (1|problem),
    family = binomial("logit"),
    data = dat_Hother[dat_Hother$culture == "Collectivistic", ],

```

```
control = glmerControl(optimizer = "bobyqa",
  optCtrl = list(maxfun = 1e5)))

summary(m_c)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: corrected ~ feedback * justification + (feedback * justification ||
Country) + (1 | problem)
Data: dat_Hother[dat_Hother$culture == "Collectivistic", ]
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik  | deviance | df.resid |
|---------|---------|---------|----------|----------|
| 17889.5 | 17957.5 | -8935.8 | 17871.5  | 14069    |

Scaled residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -2.2695 | -0.8405 | -0.5039 | 0.9467 | 2.8886 |

Random effects:

| Groups    | Name                   | Variance | Std.Dev. |
|-----------|------------------------|----------|----------|
| Country   | (Intercept)            | 0.267211 | 0.51692  |
| Country.1 | feedback               | 0.005844 | 0.07645  |
| Country.2 | justification          | 0.028916 | 0.17005  |
| Country.3 | feedback:justification | 0.053052 | 0.23033  |
| problem   | (Intercept)            | 0.244702 | 0.49467  |

Number of obs: 14078, groups: Country, 22; problem, 6

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )     |
|------------------------|----------|------------|---------|--------------|
| (Intercept)            | -0.81854 | 0.23512    | -3.481  | 0.000499 *** |
| feedback               | 0.63729  | 0.05961    | 10.690  | < 2e-16 ***  |
| justification          | 0.37731  | 0.07071    | 5.336   | 9.51e-08 *** |
| feedback:justification | -0.15924 | 0.09719    | -1.638  | 0.101333     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    | -0.125 |        |        |
| justificatn | -0.111 | 0.390  |        |
| fdbck:jstfc | 0.069  | -0.539 | -0.483 |

Test pre-registered hypotheses

```
alpha <- 0.05

# -----
# H2

# feedback increases correction rate
```

```

# One-tailed test for H2a
# Testing if the coefficient for 'feedback' is significantly greater than 0
b_H2a <- fixef(m_all)["feedback"]
SE_H2a <- sqrt(diag(vcov(m_all))["feedback"])
z_H2a <- b_H2a / SE_H2a
p_value_H2a <- 1 - pnorm(z_H2a) # right-tailed test
H2a <- ifelse(p_value_H2a < alpha, 1, 0)

# One-tailed tests for H2b
models <- list(m_i, m_c)
names <- c("Individualistic", "Collectivistic")
H2b_pvals <- numeric(length(models))
H2b_results <- numeric(length(models))

for (i in 1:length(models)) {
  b <- fixef(models[[i]])["feedback"]
  SE <- sqrt(diag(vcov(models[[i]]))["feedback"])
  z <- b / SE
  p_value <- 1 - pnorm(z) # right-tailed test
  H2b_pvals[i] <- p_value
  H2b_results[i] <- ifelse(p_value < alpha, 1, 0)
}

names(H2b_pvals) <- names
names(H2b_results) <- names

```

### H3

```
# -----
# H3

# justification increases correction rate

# One-tailed test for H3a
# Testing if the coefficient for 'justification' is significantly greater than 0
b_H3a <- fixef(m_all)["justification"]
SE_H3a <- sqrt(diag(vcov(m_all))["justification"])
z_H3a <- b_H3a / SE_H3a
p_value_H3a <- 1 - pnorm(z_H3a) # right-tailed test
H3a <- ifelse(p_value_H3a < alpha, 1, 0)

# One-tailed tests for H3b
models <- list(m_i, m_c)
names <- c("Individualistic", "Collectivistic")
H3b_pvals <- numeric(length(models))
H3b_results <- numeric(length(models))

for (i in 1:length(models)) {
  b <- fixef(models[[i]])["justification"]
  SE <- sqrt(diag(vcov(models[[i]]))["justification"])
  z <- b / SE
  p_value <- 1 - pnorm(z) # right-tailed test
  H3b_pvals[i] <- p_value
  H3b_results[i] <- ifelse(p_value < alpha, 1, 0)
}

names(H3b_pvals) <- names
names(H3b_results) <- names
```

### Hypothesis testing summary: by-problem random intercepts

Table 12: Summary of preregistered hypothesis tests after including by-problem random intercepts

| Hypothesis | Prediction                                                                             | Supported | p-value(s)                                        |
|------------|----------------------------------------------------------------------------------------|-----------|---------------------------------------------------|
| H1a        | Less than half of correct answers will arise from error correction during deliberation | No        | .875                                              |
| H1b        | Baseline performance is below 50% in both cultural groups                              | No        | Individualistic = .984; Collectivistic = .304     |
| H2a        | Feedback increases correction overall                                                  | Yes       | < .001                                            |
| H2b        | Feedback increases correction in both cultural groups                                  | Yes       | Individualistic = < .001; Collectivistic = < .001 |
| H3a        | Justification increases correction overall                                             | Yes       | < .001                                            |
| H3b        | Justification increases correction in both cultural groups                             | Yes       | Individualistic = < .001; Collectivistic = < .001 |

## Model-based maps

### Maps for H1 variable

These maps are based on the models with by-problem random intercepts included

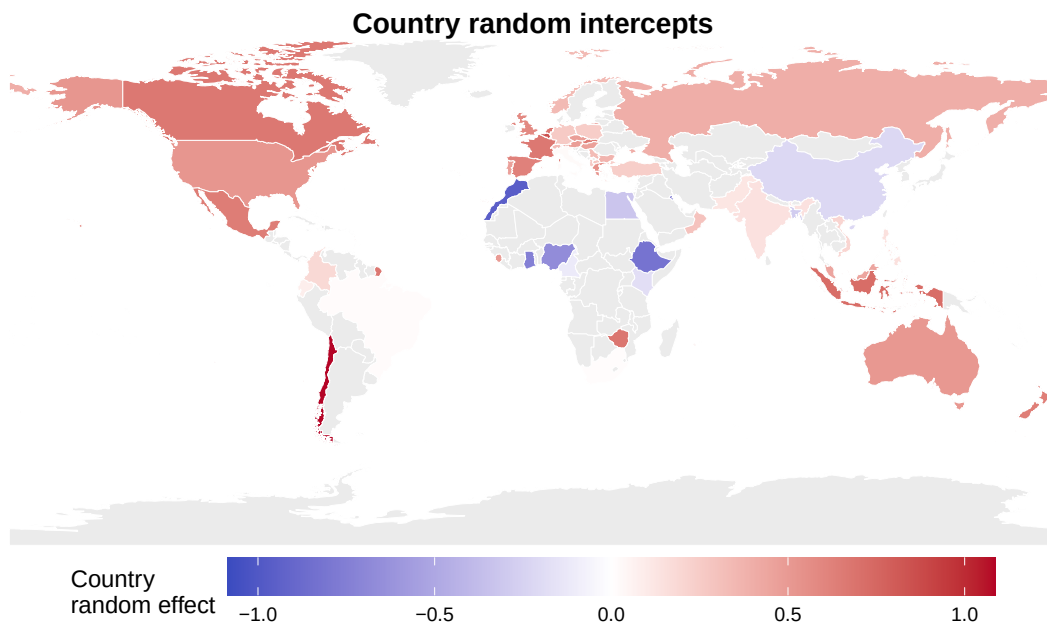


Figure 9: By-country random intercepts map, extracted from model predicting the fraction of correct choices made in the reflexive phase (same as H1). Note that for this plot the fixed-effect was added to the intercept, hence countries with negative values (blue colors) are countries in which participants, in the baseline condition, tended to make the majority of correct responses in the intuitive phase.

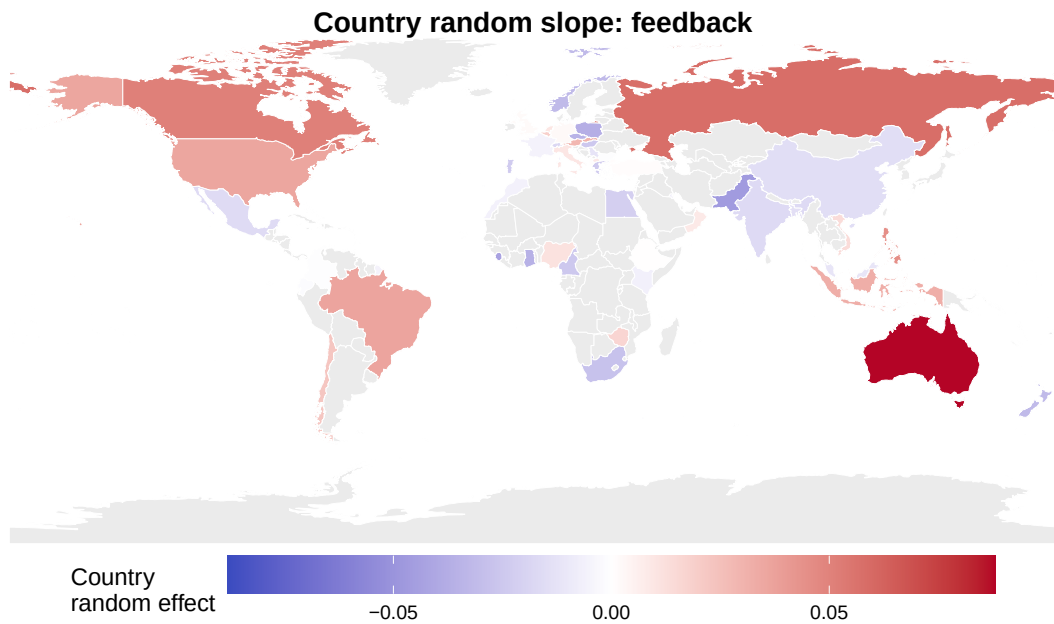


Figure 10: By-country random slope map, extracted from model predicting the fraction of correct choices made in the reflexive phase (same as H1). This maps shows country-specific variations from the overall feedback effect.



Figure 11: By-country random slope map, extracted from model predicting the fraction of correct choices made in the reflexive phase (same as H1). This maps shows country-specific variations from the overall justification effect.

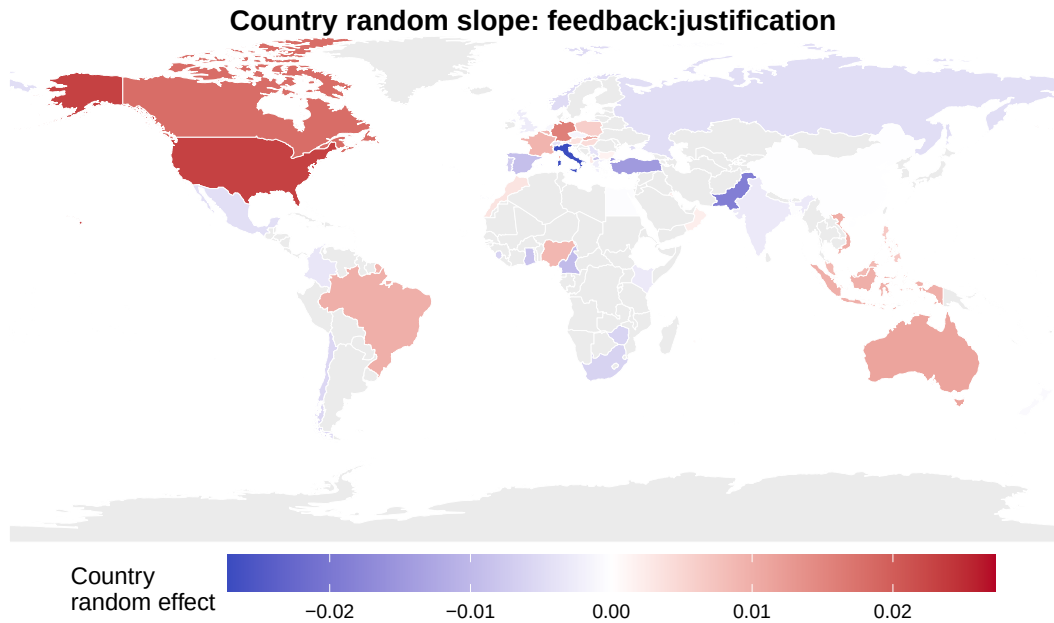


Figure 12: By-country random slope map, extracted from model predicting the fraction of correct choices made in the reflexive phase (same as H1). This map shows country-specific variations from the overall combined feedback + justification effect.

### **Maps for H2 and H3 variable**

These maps are based on the models with by-problem random intercepts included

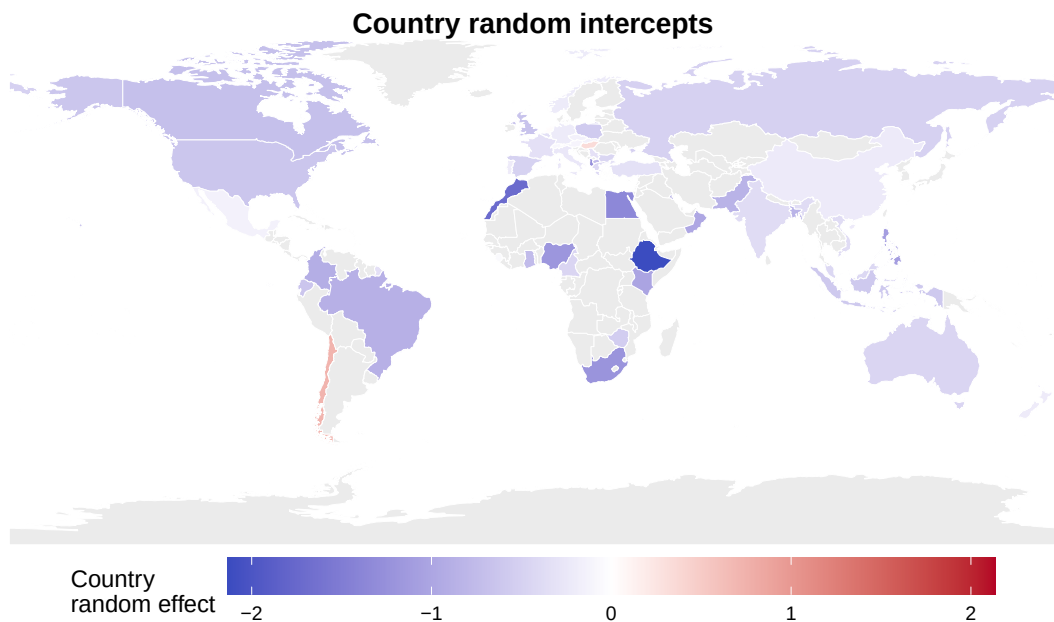


Figure 13: By-country random intercepts map, extracted from model predicting the probability that an error in the intuitive phase is corrected in the reflexive phase (variable for H2 and H3). Note that for this plot the fixed-effect was added to the intercept, hence countries with negative values (blue colors) are countries in which participants, in the baseline condition, tended to make the majority of correct responses in the intuitive phase.

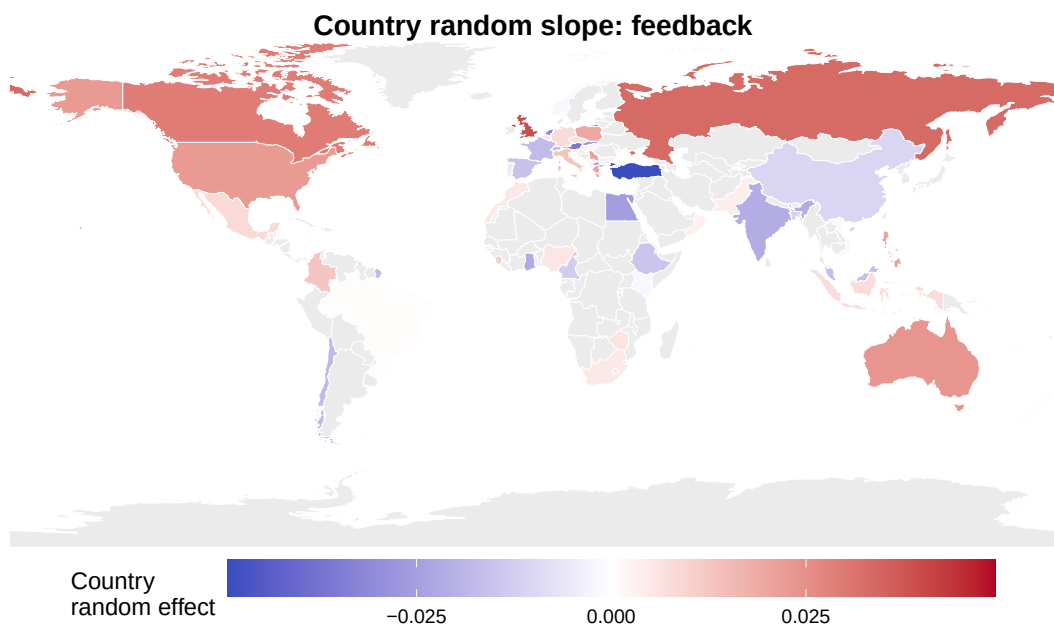


Figure 14: By-country random intercepts map, extracted from model predicting the probability that an error in the intuitive phase is corrected in the reflexive phase (variable for H2 and H3). This map shows country-specific variations from the overall feedback effect.

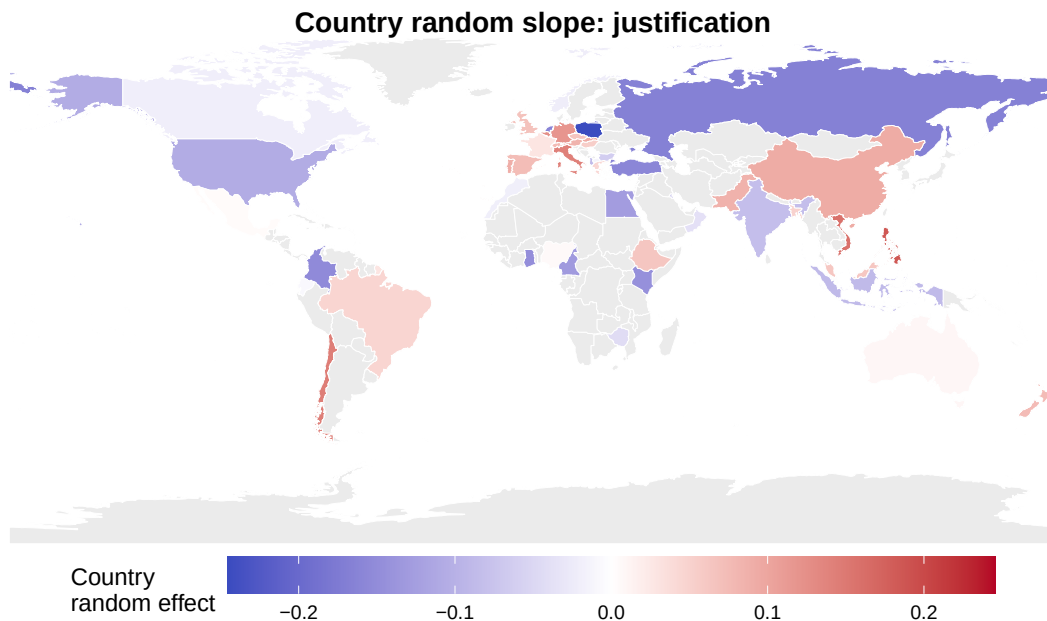


Figure 15: By-country random intercepts map, extracted from model predicting the probability that an error in the intuitive phase is corrected in the reflexive phase (variable for H2 and H3). This map shows country-specific variations from the overall justification effect.

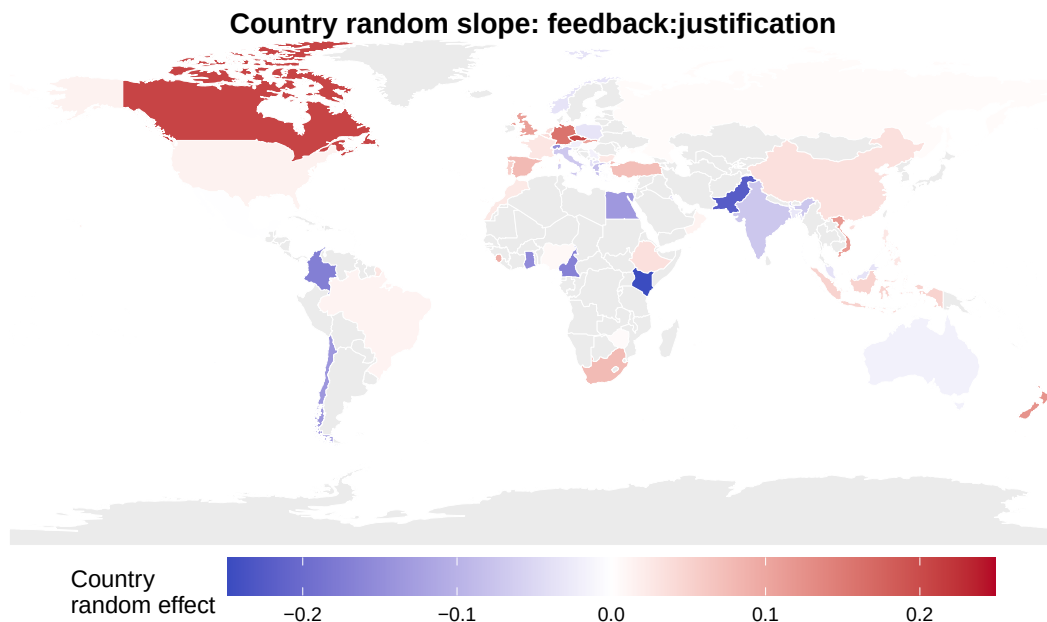


Figure 16: By-country random intercepts map, extracted from model predicting the probability that an error in the intuitive phase is corrected in the reflexive phase (variable for H2 and H3). This map shows country-specific variations from the overall combined feedback + justification effect.

## By-problem & by-participant random intercepts

As there were different participants in different groups, once I enter participant as grouping factor of random effects the variance associated with country-specific random slopes goes to zero (as variability is better accounted for as individual differences) and the model return singular fit. However, the below model with 3 crossed random effects fits well

```
mH1_all <- glmer(correct_phase2 ~ feedback * justification +
  (1|Country)+(1|problem) + (1|participant),
  family = binomial("logit"), data = dat_H1,
  control = glmerControl(optimizer = "bobyqa",
    optCtrl = list(maxfun = 1e5)))

summary(mH1_all)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: correct_phase2 ~ feedback * justification + (1 | Country) + (1 |
  problem) + (1 | participant)
Data: dat_H1
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
```

| AIC     | BIC     | logLik   | deviance | df.resid |
|---------|---------|----------|----------|----------|
| 30322.6 | 30379.4 | -15154.3 | 30308.6  | 24583    |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.6319 | -0.9523 | 0.5144 | 0.6962 | 2.1793 |

Random effects:

| Groups      | Name        | Variance | Std.Dev. |
|-------------|-------------|----------|----------|
| participant | (Intercept) | 0.3657   | 0.6047   |
| Country     | (Intercept) | 0.2516   | 0.5016   |
| problem     | (Intercept) | 0.2543   | 0.5042   |

Number of obs: 24590, groups: participant, 11038; Country, 53; problem, 6

Fixed effects:

|                        | Estimate | Std. Error | z value | Pr(> z )   |
|------------------------|----------|------------|---------|------------|
| (Intercept)            | 0.24851  | 0.22030    | 1.128   | 0.259      |
| feedback               | 0.39653  | 0.04464    | 8.884   | <2e-16 *** |
| justification          | 0.26481  | 0.04591    | 5.769   | 8e-09 ***  |
| feedback:justification | -0.06031 | 0.06278    | -0.961  | 0.337      |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | fedbck | jstfct |
|-------------|--------|--------|--------|
| feedback    | -0.108 |        |        |
| justificatn | -0.105 | 0.524  |        |
| fdbck:jstfc | 0.077  | -0.706 | -0.729 |